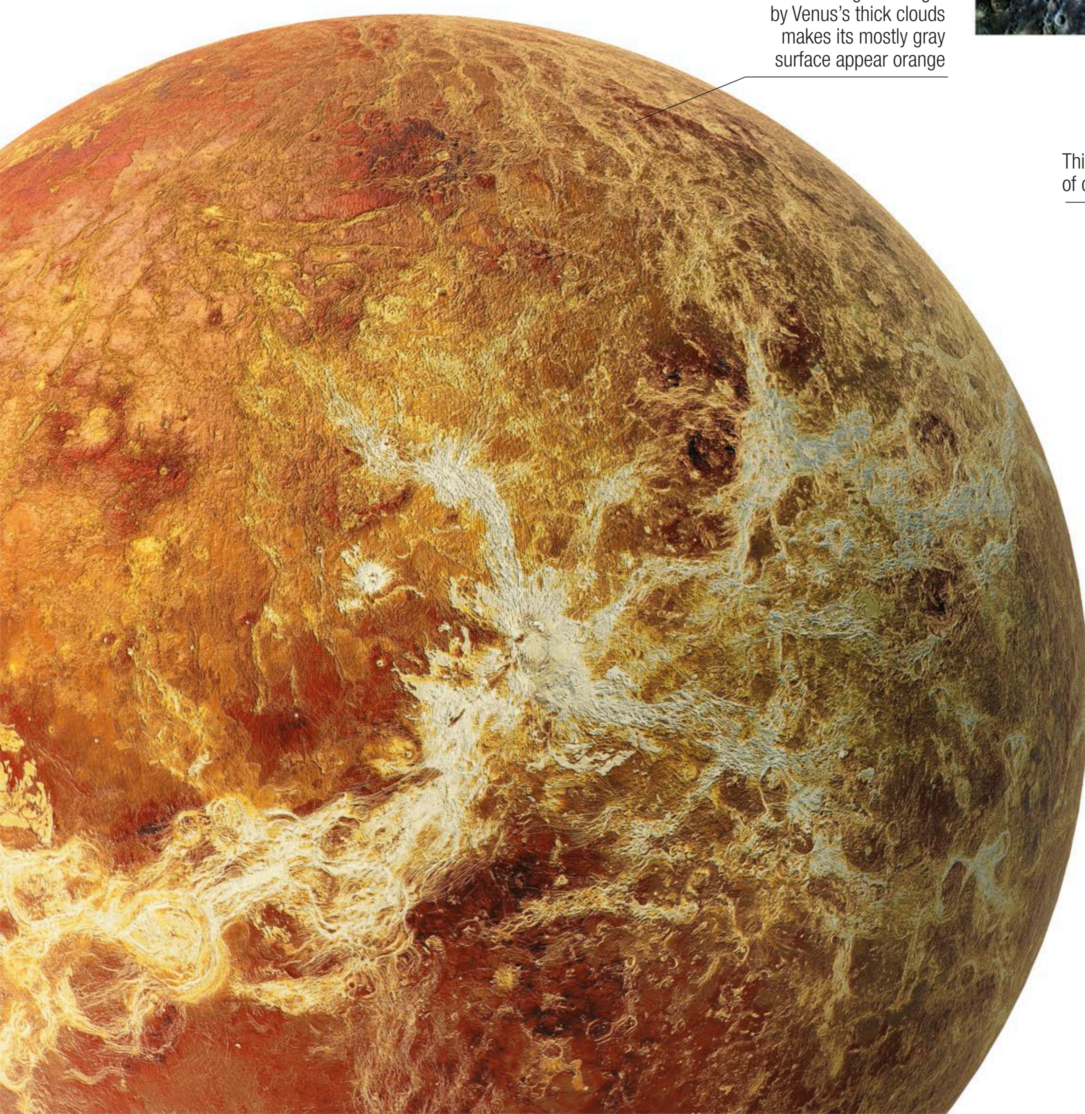
THE INNER PLANETS

THE INNER PLANETS, INCLUDING OUR OWN PLANET, ARE ALL ROCKY BODIES. BASKING IN THE WARMTH RELATIVELY CLOSE TO THE SUN, ALL THESE FOUR BODIES HAVE SOLID, ROCKY CRUSTS, BUT ARE DIVERSE WORLDS.

The sizes of the planets' atmospheres are controlled by their gravitational pull. Mercury is too small to sustain any significant atmosphere. Mars's air is gradually being lost, and was previously be much more extensive. Venus and Earth can retain thick atmospheres.

Venus

Despite almost being Earth's twin in terms of size, with a diameter of 7,521 miles (12,104 km), Venus has evolved along a very different path to our planet. The planet rotates very slowly every 243 Earth days and is covered in outflows from many volcanoes. Its extremely thick atmosphere gives it a surface pressure 90 times higher than Earth's, at a temperature of around 860°F (460°C).



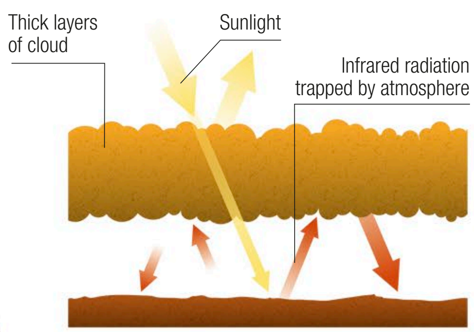
The filtering of sunlight by Venus's thick clouds makes its mostly gray surface appear orange

Mercury

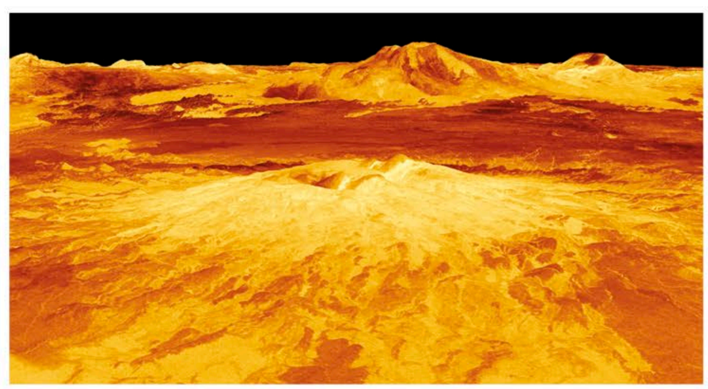
With a diameter of 3,032 miles (4,879 km), Mercury is the smallest of the terrestrial planets. It is, however, very dense. It has a huge iron core that generates a planet-wide magnetic field, similar to that of Earth. The planet has an extremely thin atmosphere that barely differs from a vacuum, much of which is kicked up by particles from the Sun striking the surface. It is a world of temperature extremes, extending from -274 to 788° F (-170 to 420° C).

Mercury's surface is similar to the Moon in appearance: covered in countless impact craters, but also some smooth areas formed from lava flows.

◁ **Mercury crater**
Fresh impact craters like this one show that Mercury's soil and rocks are varied in composition. This strike by a small asteroid has thrown up bright material from just under the surface.



◁ **The greenhouse effect**
The extremely high surface temperature on Venus is due to its thick atmosphere and clouds. The gases surrounding the planet allow in some sunlight, which heats the planet's surface. The warmed ground glows at infrared wavelengths. The atmosphere prevents this heat from escaping into space.



△ **Venus surface**
Radar observations that can penetrate Venus's thick clouds reveal its surface to be almost entirely covered by volcanoes and lava flows; there are very few impact craters.